

Multiple Choice

1. **Answer:** C

Options: A) tying back long hair.; B) wearing safety goggles.; C) performing each step quickly.; D) wiping up spills immediately.

Note: Performing each step quickly compromises safety by increasing the risk of accidents and errors, whereas safe laboratory practices emphasize careful and deliberate actions to ensure a secure working environment.

2. **Answer:** B

Options: A) They release oxygen into the air that animals breathe.; B) They put nutrients into the soil that plants use to grow.; C) They provide shelter in forests where animals can hide.; D) They use sunlight to make food for plants and animals.

Note: Decomposers break down dead organic matter, returning essential nutrients to the soil, which plants absorb to support their growth and, in turn, provide food and oxygen for other organisms in the ecosystem.

3. **Answer:** C

Options: A) the lack of water; B) the presence of igneous rock; C) the very thin atmosphere; D) the lack of magnetic poles

Note: The lack of weather on the Moon is primarily due to its very thin atmosphere, which is insufficient to support weather phenomena like clouds, rain, or wind that are driven by changes in air pressure and temperature on Earth.

4. **Answer:** A

Options: A) a filter; B) a measuring cup; C) a balance; D) a magnifying glass

Note: A filter is the best tool for separating soil from water because it allows the liquid to pass through while trapping the solid particles, effectively separating the two components of the mixture.

5. **Answer:** D

Options: A) erosion of rocks; B) violent earthquakes; C) volcanic eruption; D) movement of plates

Note: The movement of tectonic plates is the key process that leads to the formation of mountain ranges, as the collision, subduction, or sliding of these plates results in the elevation and deformation of the Earth's crust.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: A human sperm cell contains 23 chromosomes, which is half the number found in a typical human cell that has 46 chromosomes.

7. **Answer:** True

Options: A) True; B) False

Note: The speedometer measures the rotational speed of the vehicle's wheels and converts that information into a display of the vehicle's current speed, providing essential feedback to the driver.

8. **Answer:** True

Options: A) True; B) False

Note: Sexual reproduction in daffodils leads to the combination of genetic material from two parent plants, resulting in offspring that have varied inherited traits and increased genetic diversity within the population.

9. **Answer:** True

Options: A) True; B) False

Note: Overgrazing by an excessive number of bison can lead to the depletion of grass, resulting in less ground cover and increased bare soil as the vegetation is unable to recover from the intense grazing pressure.

10. **Answer:** True

Options: A) True; B) False

Note: During the winter months, the tilt of the Earth's axis causes North America to receive sunlight at a lower angle and for fewer hours each day compared to the summer months, resulting in less direct sunlight.

Long Form Response

11. **Answer:** The body's regulatory systems, particularly the hypothalamus, play a crucial

role in maintaining a normal temperature of 98.6° Fahrenheit. *When the body experiences changes in the environment,*

Note: correct because it accurately identifies the hypothalamus as the central regulator of body temperature, such as sweating and shivering—as well as the role of blood vessel adjustments in maintaining a stable internal temperature.

11. **Answer:** The visual properties of minerals, such as color and luster, play a crucial role in their identification. Color can often provide a quick clue; for example, the bright yellow of sulfur or the deep blue of lapis lazuli can help distinguish these minerals from others. Luster describes how light interacts with a mineral's surface, ranging from metallic to glassy or dull. For instance, pyrite has a metallic luster that sets it apart from similar-looking minerals, while quartz, with its vitreous luster, is easily recognized. By combining observations of color and luster, scientists can effectively narrow down the possibilities and identify minerals more accurately.

Note: correct because it effectively illustrates how specific visual properties, namely color and luster, serve as distinguishing features for identifying minerals, using relevant examples to support the analysis.

End of Answer Key